

# diff - diff

Callaway-Sant'Anna. Borusyak-Jaravel-Spiess.

Cengiz-style stacking. The classic 2x2.

One regression  
to rule them all.

LP-DiD: Local Projections DiD.

*Now in diff-diff.*

*Dube, Girardi, Jordà & Taylor (2025).*

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# The staggered revolution left you with a shelf.

## Classic 2x2 DiD

Where you started.

One clean comparison.

## Callaway-Sant'Anna

Group-time ATTs.

Its own aggregation layer.

## Stacked DiD (Cengiz et al.)

Dataset duplication.

Its own weighting scheme.

## Borusyak-Jaravel-Spiess

Imputation.

Its own two-step machinery.

*Three fixes for TWFE's negative weights - plus the textbook case they outgrew.*

**Same causal question. Four mental models.**

# Same question.

## Four different machines.

### Staggered State Policy

Minimum wage rolls out state by state. CS? Stacked? Which aggregation?

### Phased Product Rollout

The feature ships to markets in waves. Event study? Which estimator?

### Programs That Switch Off

The retention treatment ends. Absorbing-only estimators don't apply.

### Survey-Weighted Panels

CPS-style pweights, strata, PSUs. Which estimator takes a design?

# LP-DiD.

*It's just a regression.*

A local projection per horizon: long difference,  
time fixed effects, clean controls. Plain OLS.

## One long difference per horizon.

OLS on clean controls. You can read every comparison it makes.

## Weights provably non-negative.

Clean-control baseline result (Eqs. 9-10). With covariates, prefer RA.

## The estimand is a dial.

Variance-weighted for precision, or equal-weighted - your call.

*Local projections (Jordà 2005), pointed at DiD.*

# Clean controls, horizon by horizon.



Each horizon is one regression you could run by hand.

*Already-treated units never serve as controls.*

# The zoo was one regression all along.

$$y_{i,t+h} - y_{i,t-1} = \beta_h \Delta D_{it} + \delta_t^h + e_{it}^h$$

$$E(\beta_h) = \sum_{g \neq 0} \omega_{g,h} \tau_h^g, \quad \omega_{g,h} \geq 0$$

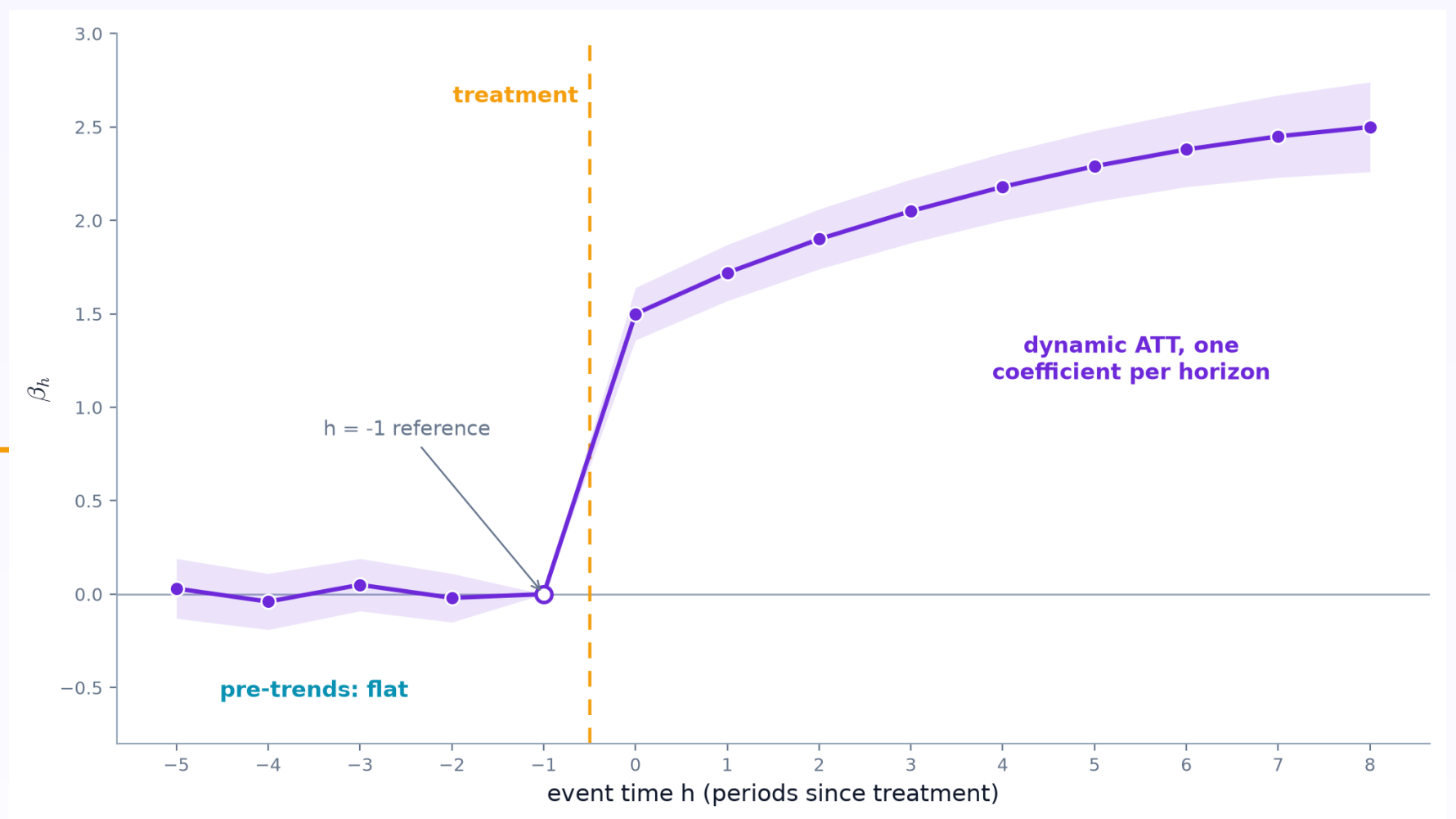
always non-negative -- no forbidden comparisons



|                                          |                                 |                                        |
|------------------------------------------|---------------------------------|----------------------------------------|
| <code>reweight=True</code>               | <b>== Callaway-Sant'Anna</b>    | <i>exact (Sec. 3.7)</i>                |
| <code>default (variance-weighted)</code> | <b>== Cengiz et al. stacked</b> | <i>exact (Sec. 3.7)</i>                |
| <code>h=0, single cohort</code>          | <b>== classic 2x2 DiD</b>       | <i>exact (Sec. 2.2)</i>                |
| <code>pmd= 'max '</code>                 | <b>-&gt; BJS imputation</b>     | <i>exact single-cohort (fn. 10-11)</i> |

*Numerical equivalences proven in the paper - not analogies.*

# Pre-trends are just negative horizons.



The event study is native - not a post-hoc aggregation.

# The Code.

Same sklearn-like API as every diff-diff estimator.

```
from diff_diff import LPDiD

result = LPDiD(
    pre_window=5, post_window=10,
).fit(
    data,
    outcome='revenue', unit='store',
    time='week', treatment='treated',
)

print(result.att)          # pooled post ATT = 2.31
result.event_study         # one beta per horizon

# flip one switch:
#   reweight=True   -> Callaway-Sant'Anna
#   non_absorbing=  -> on/off treatments (entry effects)
#   survey_design=  -> pweights + strata + PSUs (default path)
```

Callaway-Sant'Anna is literally a keyword argument.



# Production-ready.

## Cluster-Robust SEs

Unit-level by default,  
matches Stata `lstdid`

## Covariates, Two Ways

RA path (BJS-style) or  
direct with guardrails

## PMD Baselines

Premean differencing +  
pooled pre/post ATTs

## Non-Absorbing Treatment

First-entry & effect-  
stabilization (Eq. 12/13)

## Survey Designs

`pweights`, `strata`, `PSU`, `FPC`;  
Binder TSL (default path)

## Composition Control

`no_composition` fixes the  
post-window sample

*Composable where validated - `survey_design` runs the default path only.*

# Validated.

*Against the authors' own tooling - and against our own shelf.*

## Authors' R recipes (danielegirardi/lpdid)

Event-study and pooled estimands match to  $\sim 1e-13$ .

## Independent fixest reconstruction

Non-absorbing Eq. 12/13, variance-weighted: point + SE to  $\sim 1e-13$ .

## survey::svyglm, end to end

Default-path survey: per-horizon point, SE, and df all pinned.

## Equivalences tested, not cited

reweight == our CallawaySantAnna; PMD == our ImputationDiD (1 cohort).

*Now in diff-diff.*

# LP-DiD.

```
$ pip install --upgrade diff-diff
```

[github.com/igerber/diff-diff](https://github.com/igerber/diff-diff)

## diff - diff

Difference-in-Differences for Python